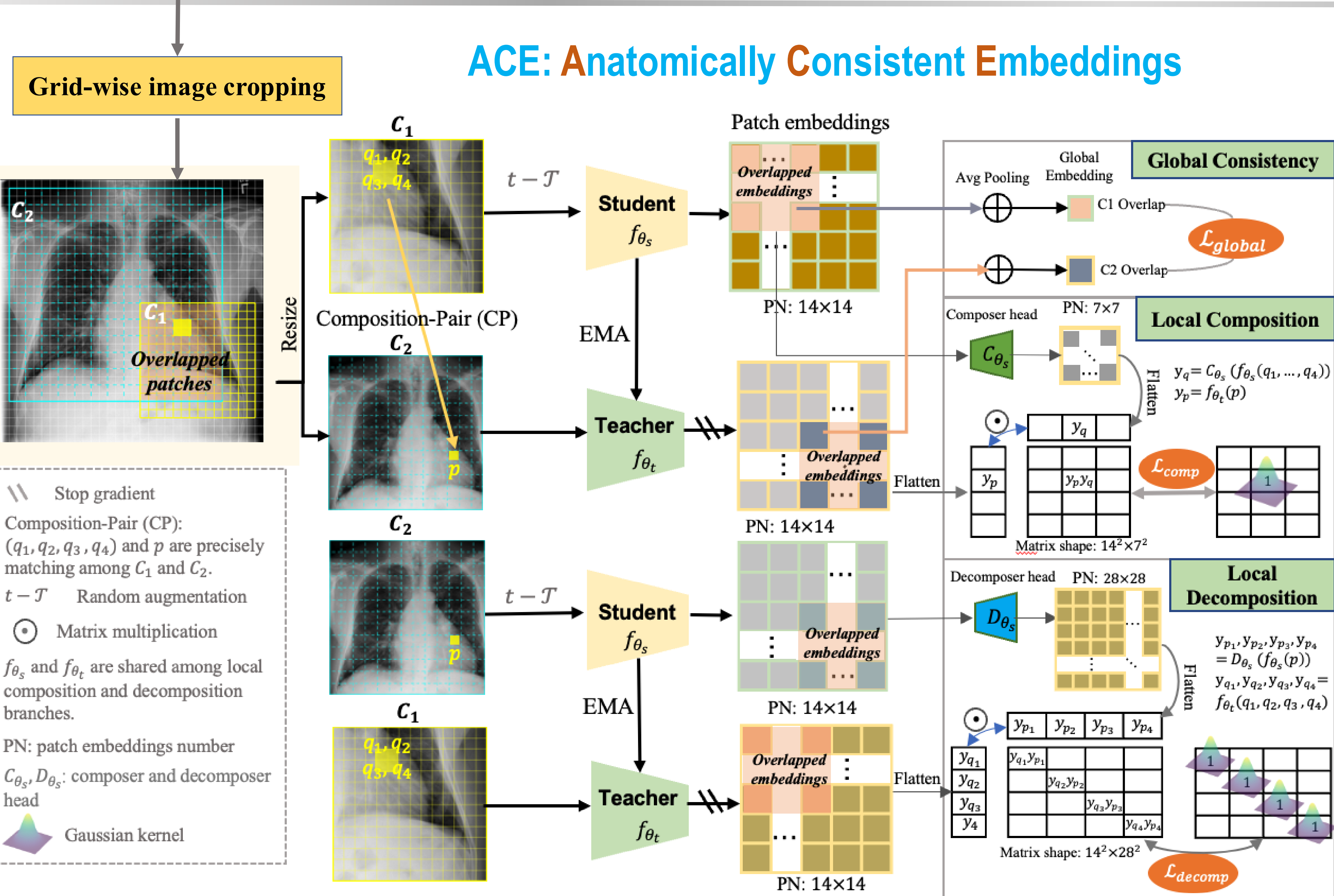
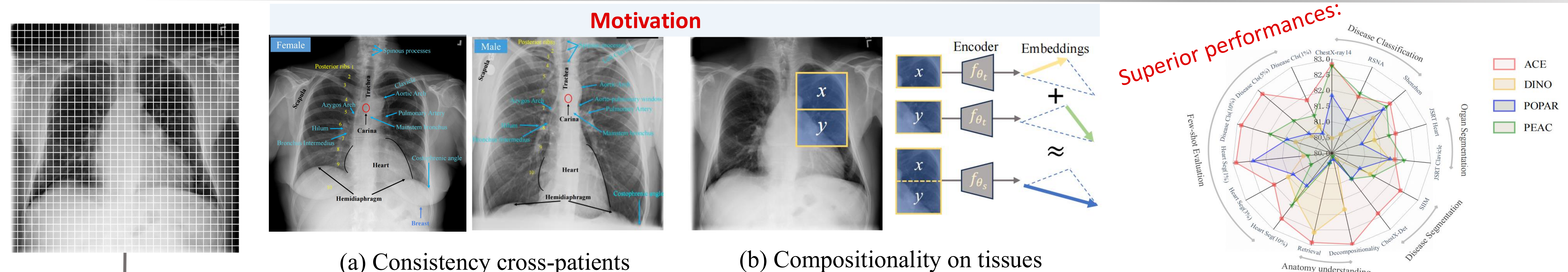




Background: Self-supervised learning (SSL) pretrains generic source models without using expert annotation, allowing the pretrained models to be uncovered of learned/emergent properties and quickly fine-tuned into high-performance application-specific target models and minimizing annotation cost.

Motivation: From an anatomical point of view, medical images acquired from the same image protocol are high consistency and exhibit decomposability of organs and tissues. This paper introduces a novel SSL approach, called ACE (anatomically consistent embeddings in composition and decomposition).

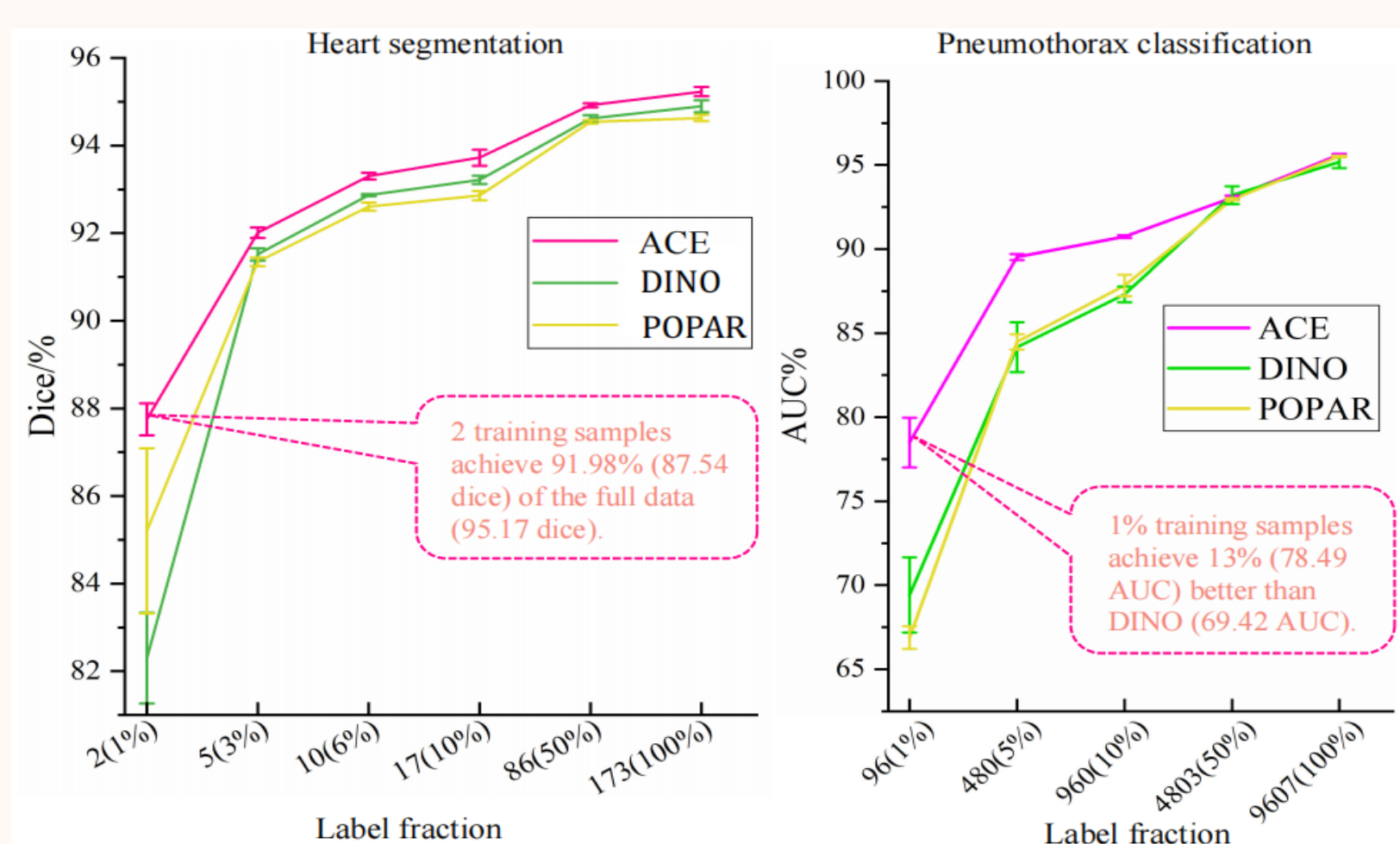
Contributions: We propose to learn composition and decomposition via stable grid-based matrix matching, demonstrate that (1) ACE achieves significantly better performance for transfer learning than the existing state-of-the-art fully/self-supervised methods; (2) ACE's novel learned and emergent properties including image retrieval, cross-patient anatomy correspondence, locality and symmetry of anatomical structures.



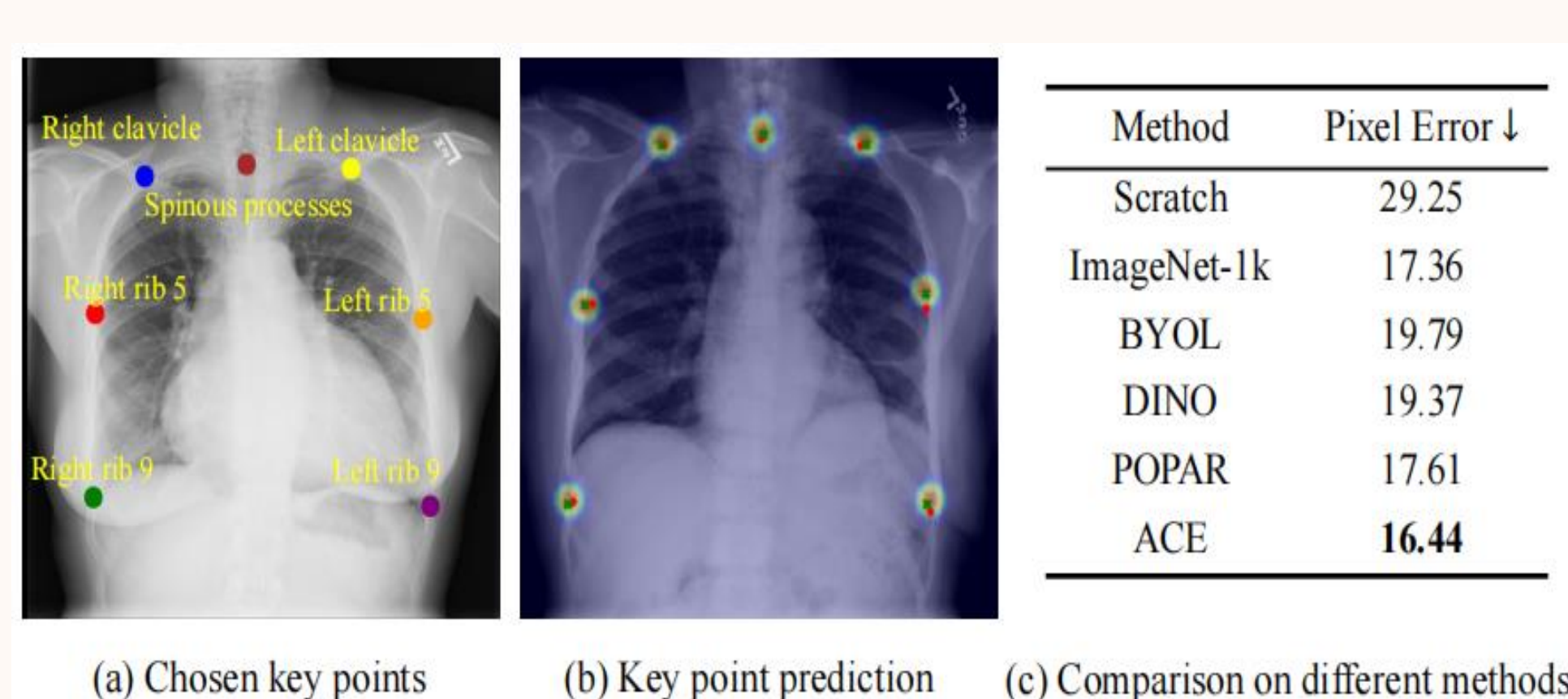
1. built on **Student-Teacher Model** and update Teacher model by EMA;
2. **global consistency**: learns coarse-grained semantic features from the **overlapped regions**;
3. **local consistency**: learns fine-grained local patterns via **composition and decomposition**.

Downstream transferability of ACE

Downstream 1: Data efficiency evaluation



Downstream 2: Key-point detection

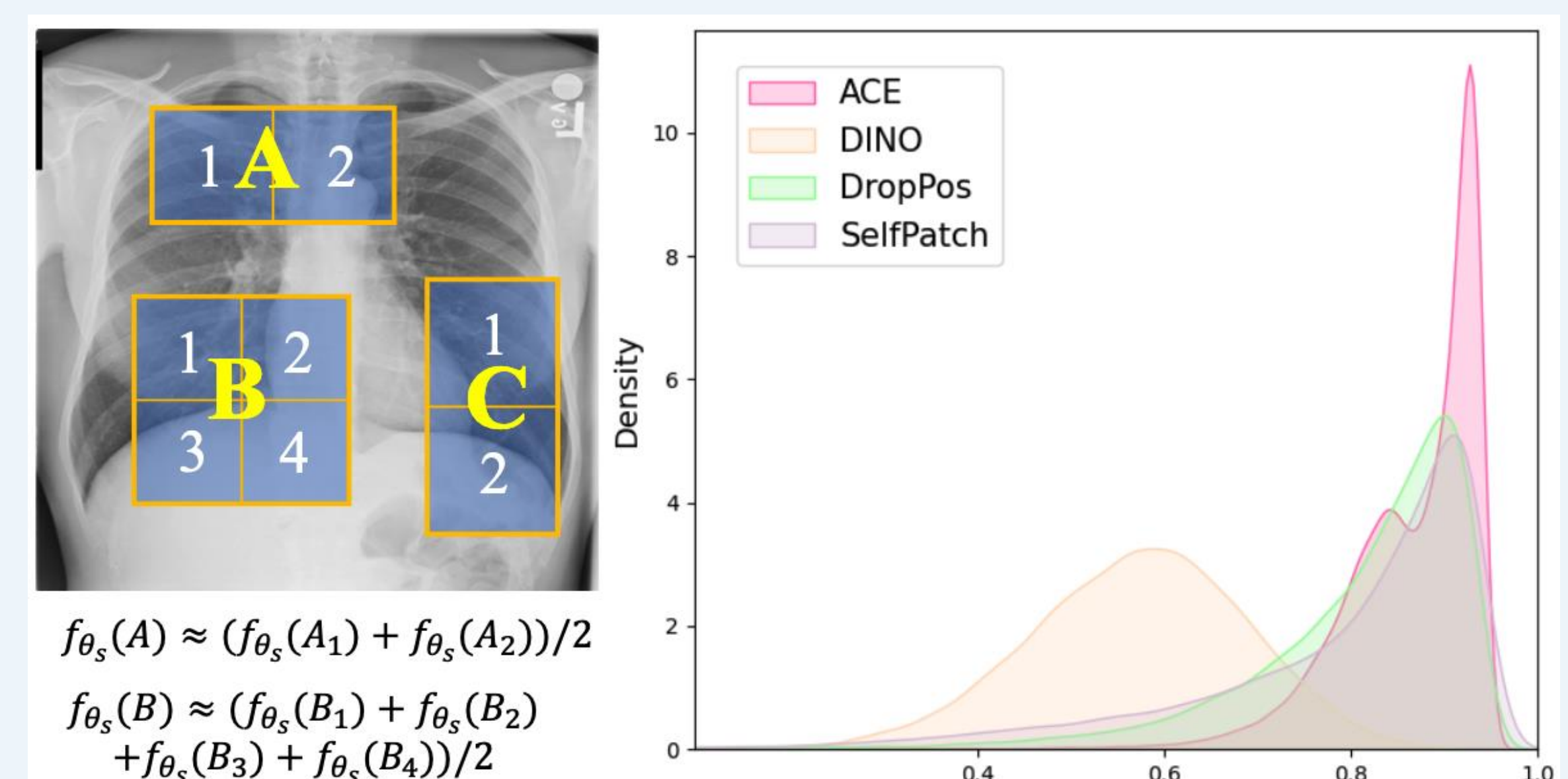


Downstream 3: Fine-tuning evaluation

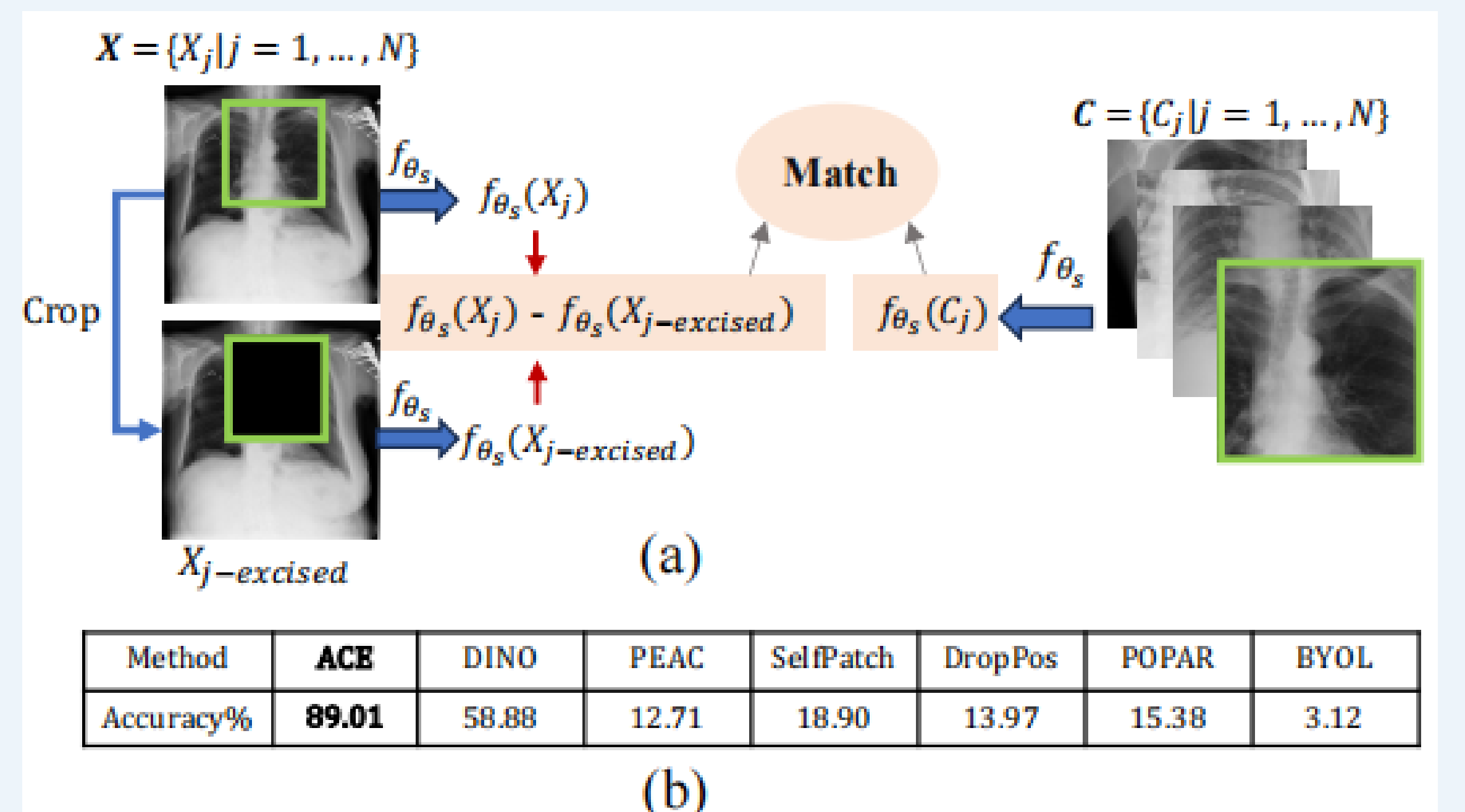
Pretraining Methods	Backbone	Classification			Segmentation				
		ChestX-ray14	ShenZhen	RSNA Pneumonia	JSRT clavicle	JSRT heart	ChestX-Det	SIIM	
Scratch									
-	Vit-B	71.59 ± 0.50	83.18 ± 0.55	63.17 ± 0.33	78.12 ± 0.07	93.95 ± 0.09	60.04 ± 0.41	71.01 ± 2.07	
	Swin-B	79.28 ± 0.07	84.61 ± 0.81	66.06 ± 0.01	87.13 ± 1.95	92.41 ± 1.27	64.52 ± 0.24	73.69 ± 0.66	
Global features									
DINO [6]	Vit-B	78.93 ± 0.57	94.79 ± 0.47	71.08 ± 0.19	83.04 ± 0.72	94.77 ± 0.39	69.30 ± 0.28	72.52 ± 2.29	
BYOL [12]	Swin-B	79.44 ± 0.55	89.74 ± 2.84	73.43 ± 0.20	90.14 ± 0.42	94.80 ± 0.11	72.56 ± 0.36	77.04 ± 0.69	
DINO [6]	Swin-B	80.48 ± 1.13	95.61 ± 0.32	73.44 ± 0.38	91.74 ± 0.06	94.90 ± 0.11	72.25 ± 1.50	78.89 ± 0.50	
Local features									
SelfPatch [55]	Vit-B	79.27 ± 0.81	95.42 ± 0.84	70.94 ± 1.03	83.18 ± 0.96	94.78 ± 0.34	70.14 ± 0.48	73.69 ± 1.25	
Inherent structural patterns and anatomy									
Adam [23]	ResNet50	81.81 ± 0.14	96.82 ± 0.50	72.97 ± 0.38	93.30 ± 0.07	95.33 ± 0.10	73.17 ± 0.11	78.44 ± 0.31	
DropPos [47]	Vit-B	79.59 ± 0.44	93.83 ± 0.22	72.70 ± 0.39	83.64 ± 2.47	94.87 ± 0.37	69.69 ± 2.03	70.52 ± 2.55	
POPAP [37]	Swin-B	81.84 ± 0.02	96.97 ± 0.37	73.78 ± 0.07	91.77 ± 0.08	94.63 ± 0.06	73.01 ± 0.39	78.65 ± 0.18	
PEAC	Vit-B	80.04 ± 0.20	96.69 ± 0.30	73.77 ± 0.39	80.59 ± 0.01	94.60 ± 0.08	67.41 ± 0.06	72.00 ± 0.90	
	Swin-B	82.78 ± 0.21	97.39 ± 0.19	74.39 ± 0.66	92.19 ± 0.04	95.21 ± 0.02	73.01 ± 0.39	78.97 ± 0.26	
Vision language pretraining									
KAD [60]	Vit-B	80.48 ± 0.19	97.86 ± 0.95	72.94 ± 0.40	81.67 ± 0.06	95.07 ± 0.12	69.9 ± 0.30	72.51 ± 1.78	
ChexZero [45]	Vit-B	78.67 ± 0.07	96.34 ± 0.13	72.70 ± 0.21	77.45 ± 0.20	93.92 ± 0.06	68.41 ± 0.37	76.41 ± 0.76	
DeViDe [34]	Vit-B	81.32 ± 0.08	98.03 ± 0.24	73.86 ± 0.42	81.73 ± 0.13	94.85 ± 0.03	70.27 ± 0.08	75.52 ± 0.46	
Ours									
ACE	Vit-B	79.84 ± 0.44	95.73 ± 0.12	71.31 ± 0.33	83.66 ± 0.56	95.03 ± 0.06	70.87 ± 0.90	74.86 ± 0.56	
	Swin-B	82.83 ± 0.08	97.87 ± 0.29	74.31 ± 0.16	91.91 ± 0.07	95.35 ± 0.06	74.37 ± 0.42	79.59 ± 0.67	

Learned properties of ACE

Learned property 1: ACE enhances feature compositionality

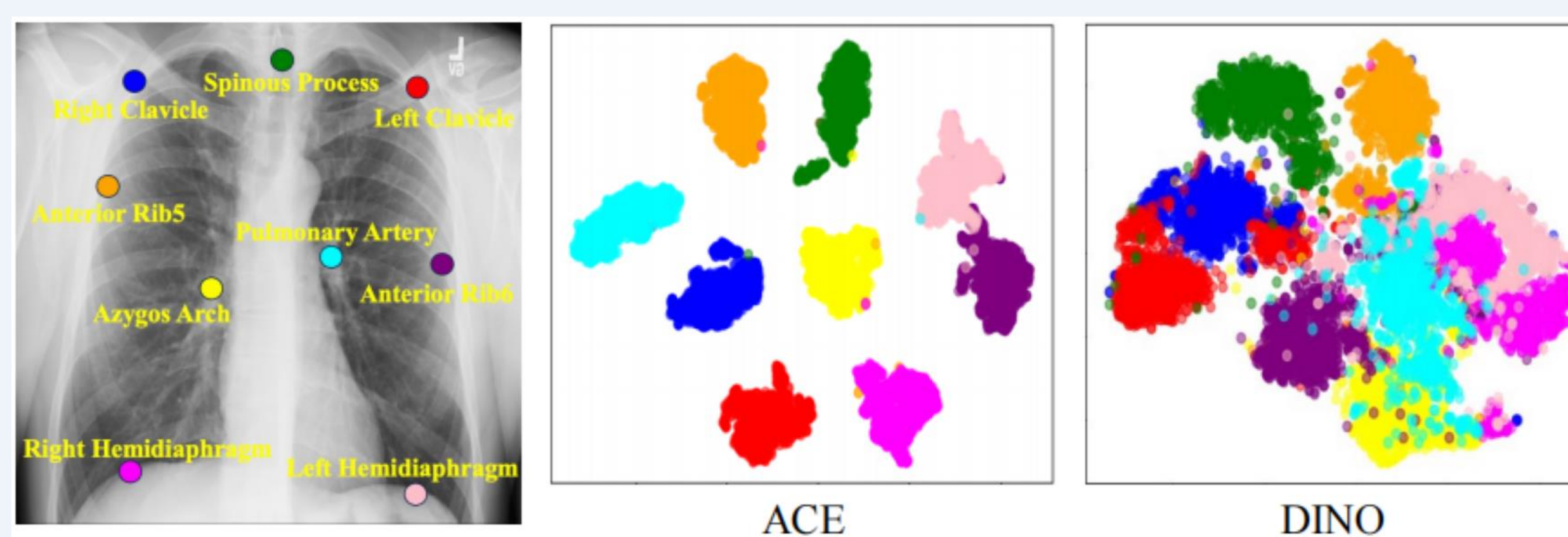


Learned property 2: ACE enhances feature decomposability



Emergent properties of ACE

Emergent property 1: ACE provides distinctive anatomical embeddings



Emergent property 2: ACE provides unsupervised cross-patient anatomy correspondence

